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SQL Project Report

HOSPITAL MANAGEMENT ANALYSIS



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SYSTEM INTRODUCTION

A hospital is more than a place where patients meet doctors. Behind every appointment, medical visit, admission, surgery, and staff shift, there is information that needs to be organized and connected.

This **Hospital Management System** is designed to represent a multi-department hospital environment through **14 interconnected tables, 93 columns, and 15,981 records**. The system brings together information about **patients, doctors, nurses, helpers, departments, wards, rooms, beds, appointments, medical records, admissions, surgeries, and staff shifts**.

OPERATIONAL CHALLENGES

Managing these activities across different areas of a hospital can become difficult when information is spread across multiple records. Management may need to quickly answer questions such as:

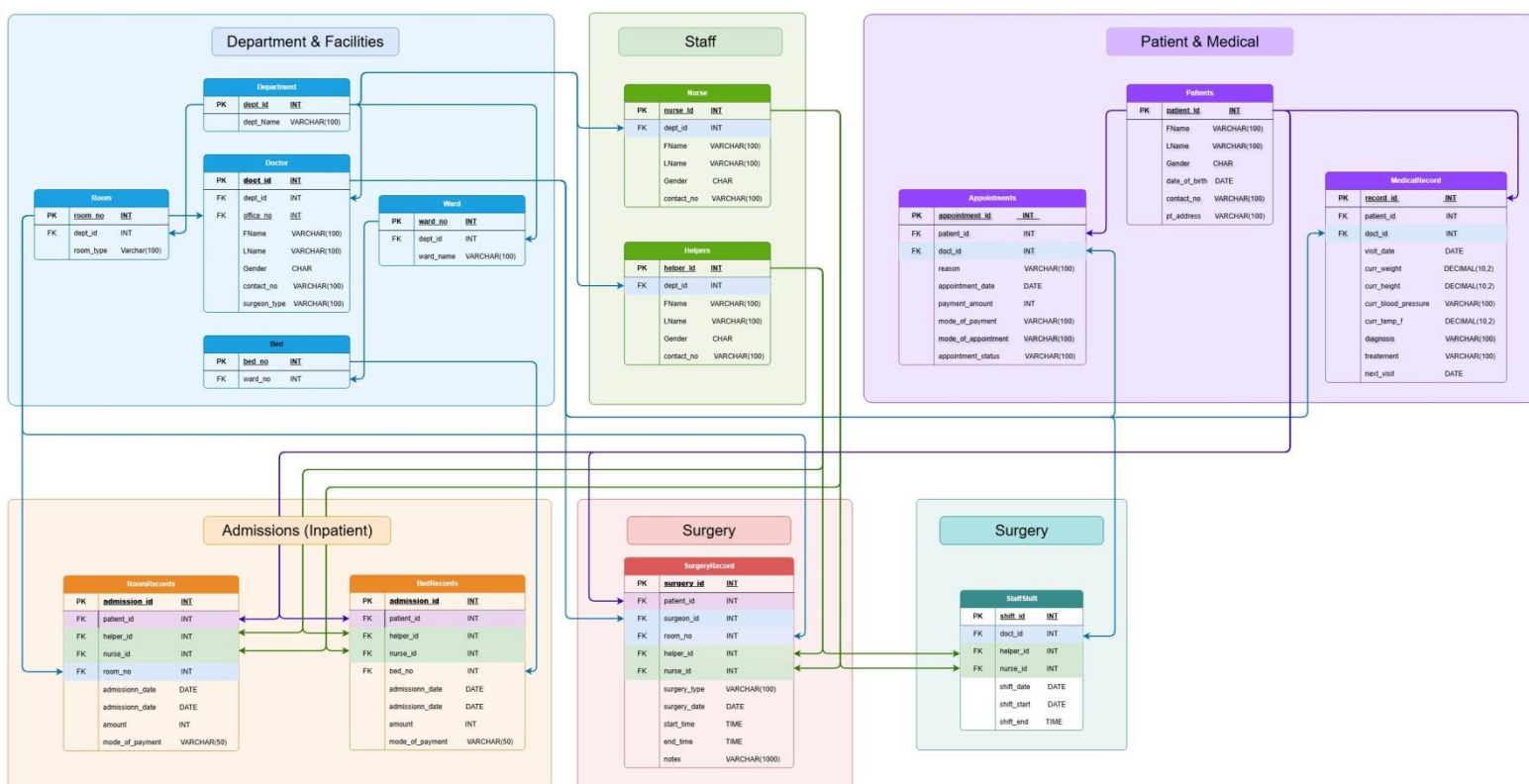
- Which patients are connected with particular appointments and doctors?
- Which beds are available when a new admission is required?
- How many appointments are expected within a department?
- How much work is being handled by individual staff members?
- Which staff members were involved in a particular surgery?
- How are admissions, payments, medical visits, and surgeries being recorded?

The main challenge is to turn the hospital's large amount of connected operational data into clear, accessible answers for management.

Table	Columns	Records	Contains
Department	2	31	Hospital departments
Room	3	530	Hospital rooms
Doctor	8	400	Doctor information

Table	Columns	Records	Contains
Nurse	6	500	Nurse information
Helpers	6	1,100	Helper information
Ward	3	63	Hospital wards
Bed	2	500	Hospital beds
Patients	7	1,500	Patient information
BedRecords	9	1,000	Bed admissions
RoomRecords	9	1,000	Room admissions
Appointment	9	1,000	Patient appointments
MedicalRecord	11	3,000	Patient medical records
StaffShift	7	2,058	Staff working shifts
SurgeryRecord	11	1,000	Surgery information

Relationship Diagram



[Diagram Link](#)

QUESTION / QUERIES / ANSWERS

Question 1

List patients with their appointment doctor and reason.

```
SELECT p.fname AS Patient,  
       d.fname AS Doctor,  
       a.reason AS Reason  
FROM patients p  
JOIN appointment a ON p.patient_Id = a.patient_Id  
JOIN doctor d ON a.doct_Id = d.doct_Id;
```

Patient	Doctor	Reason
Ilyas	Dr. Raza	Back Pain
Ubaid	Dr. Muhammad	Migraine
Ilyas	Dr. Muhammad	Diabetes
Noor	Dr. Tahir	Migraine
Sameer	Dr. Najeeb	Hypertension

Showing 5 of 1,000 results

Question 2

Show nurses who have assisted in bed admissions with patient names..

```
SELECT n.fname AS Nurse,  
       p.fname AS Patient,  
       bdr.admission_date AS Admission_Date  
FROM nurse n  
JOIN bedrecords bdr ON n.nurse_Id = bdr.nurse_Id
```

JOIN patients p **ON** p.patient_Id = bdr.patient_Id;

Nurse	Patient	Admission Date
Farah	Ali	2025-03-21
Lateef	Shahbaz	2025-04-07
Waqas	Abdul	2024-12-09
Hafsa	Neha	2025-05-16
Bushra	Sana	2025-04-26

Showing 5 of 1,000 results

Question 3

List rooms used for surgeries, the surgeon, and the surgery type.

SELECT r.room_no **AS** Room_Number,
d.fname **AS** Surgeon,
sgr.surgery_type
FROM room r
JOIN doctor d **ON** r.dept_Id = d.dept_Id
JOIN surgeryrecord sgr **ON** sgr.room_no = r.room_No;

Room no	Surgeon	Surgery Type
291	Dr. Nabeel	Otolaryngologic (ENT) Surgery
291	Dr. Rameez	Otolaryngologic (ENT) Surgery
291	Dr. Imran	Otolaryngologic (ENT) Surgery
291	Dr. Abu	Otolaryngologic (ENT) Surgery
291	Dr. Muhammad	Otolaryngologic (ENT) Surgery

Showing 5 of 38,000 results

Question 4

List each department with the number of doctors assigned to it.

```
SELECT dept.dept_name,  
       COUNT(d.doct_id) Doctors  
FROM department dept  
JOIN doctor d ON dept.dept_id = d.dept_id  
GROUP BY dept.dept_name;
```

Department	Doctors	Department	Doctors
Cardiology	7	Endocrinology	3
Neurology	5	General Medicine	52
Gastroenterology	3	Pediatrics	14
Nephrology	4	Orthopedics	5
Pulmonology	4	Gynecology	8

Showing 10 of 37 results

Question 5

Show patients who had an appointment and were admitted to a bed.

```
SELECT p.fname AS Patient,  
       d.fname AS Doctor,  
       a.appointment_Date AS Appointment_Date,  
       bdr.bed_no AS Bed_Number  
FROM patients p  
JOIN appointment a ON p.patient_id = a.patient_id  
JOIN doctor d ON d.doct_id = a.doct_id  
JOIN bedrecords bdr ON a.patient_id = bdr.patient_id;
```

Patient	Doctor	Appointment Date	Bed Number
Ilyas	Dr. Raza	2024-11-27	263
Ilyas	Dr. Muhammad	2024-09-07	470
Shehla	Dr. Raza	2024-12-31	208
Farah	Dr. Raza	2024-10-25	234
Hussain	Dr. Waqas	2025-04-18	402

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Question 6

We have a new patient for the Cardiology Ward and he/she wants a bed on a specific day. We want to find out which beds are empty in that ward on that particular day.

```

SELECT b.bed_No AS Bed_Number,
       b.ward_no AS Ward_Number,
       d.dept_Name AS Department
FROM bed b
JOIN ward w ON w.ward_No = b.ward_No
JOIN department d ON d.dept_Id = w.dept_Id
WHERE
    d.dept_Name = 'Cardiology'
    AND
    w.ward_No = 502
    AND
    b.bed_no NOT IN (
        SELECT br.bed_no
        FROM bedrecords br
        WHERE '2025-5-10' BETWEEN
            br.admission_Date AND br.discharge_Date);

```

Bed no	Ward no	Department	Bed no	Ward no	Department
16	502	Cardiology	20	502	Cardiology
17	502	Cardiology	21	502	Cardiology
18	502	Cardiology	23	502	Cardiology
19	502	Cardiology			

Question 7

There is a new virus in the city and the hospital is expecting more patients than a regular day. Management wants to see if they can manage those with the current staff or not. They want to check upcoming appointments for each department on 4 June 2025.

```
SELECT dpt.dept_name AS Department,
       a.appointment_Date AS Appointment_Date,
       COUNT(a.appointment_id) AS Appointments
FROM doctor d
JOIN department dpt ON d.dept_Id = dpt.dept_Id
JOIN appointment a ON d.doct_Id = a.doct_Id
GROUP BY
dpt.dept_name,
a.appointment_Date
HAVING a.appointment_Date = '2025-06-04';
```

Department	Date	Appointments	Department	Date	Appointments
Cardiology	2025-06-04	1	Neurology	2025-06-04	1

Question 8

A doctor is asking for a salary raise due to extra work in the previous month. Verify if he/she deserves a raise by retrieving their

- total appointments,
- total visits,
- total surgeries,
- total shifts.

```
SELECT d.fname AS Doctor,
       (
         SELECT COUNT(*) FROM appointment a
         WHERE
           a.doct_Id = d.doct_Id
           AND
           a.appointment_Date BETWEEN '2025-05-01' AND '2025-05-31'
```

```

) AS Total_Appointments,

(
SELECT COUNT(*) FROM medicalrecord mdr
WHERE
    mdr.doct_Id = d.doct_Id
    AND
    mdr.visit_Date BETWEEN '2025-05-01' AND '2025-05-31'
) AS Total_Visits,

(
SELECT COUNT(*) FROM surgeryrecord sgr
WHERE
    sgr.surgeon_Id = d.doct_Id
    AND
    sgr.surgery_Date BETWEEN '2025-05-01' AND '2025-05-31'
) AS Total_Surgeries,

(
SELECT COUNT(*) FROM staffshift stf
WHERE
    stf.doct_Id = d.doct_Id
    AND
    stf.shift_Date BETWEEN '2025-05-01' AND '2025-05-31'
) AS Total_Shifts
FROM doctor d
WHERE d.doct_Id = 1001;

```

Doctor	Total Appointments	Total Visits	Total Surgeries	Total Shifts
Dr. Najeeb	8	0	0	6

Question 9

The hospital is analyzing its daily revenue and wants to calculate the revenue generated on 10 May 2025 (including appointment revenue, room revenue, and bed revenue).

```

SELECT SUM(a.payment_amount) AS Appointment_Revenue,
       SUM(rom.amount) AS Room_Revenue,
       SUM(bed.amount) AS Bed_Revenue,

```

SUM(a.payment_amount) + **SUM**(rom.amount) + **SUM**(bed.amount) **AS**

Total_Revenue

FROM appointment a

LEFT JOIN roomrecords rom **ON** rom.admission_Date = '2025-05-10'

LEFT JOIN bedrecords bed **ON** bed.admission_Date = '2025-05-10'

WHERE

a.appointment_Date = '2025-05-10';

Appointment Revenue	Room Revenue	Bed Revenue	Total Revenue
63,936	2,464,288	1,191,012	3,719,236

Question 10

The hospital decided to give some discounts to its old customers on some services. Identify patients who have visited the hospital more than 4 times in the past year.

SELECT p.patient_Id,

p.fname **AS** Patient,

COUNT(mdr.record_id) **AS** Visits

FROM patients p

JOIN medicalrecord mdr **ON** mdr.patient_Id = p.patient_Id

WHERE mdr.visit_Date >= '2025-01-01'

GROUP BY

p.patient_id,

p.fname

HAVING COUNT(mdr.record_id) > 4;

Patient Id	Patients	Visits
483	Omar	5
1003	Farhan	5
1135	Ubaid	5

Question 11

Management received a report that a patient was given the wrong amount of anesthesia during surgery. Track which staff (surgeon, nurse, and helper) was present during the surgery of patient 967 on 16 May 2024 between 11 to 12 at night.

```
SELECT sgr.patient_id,  
       p.FName AS Patient,  
       d.FName AS Doctor,  
       n.FName AS Nurse,  
       h.FName AS Helper,  
       sgr.surgery_Type,  
       sgr.surgery_Date,  
       sgr.start_Time,  
       sgr.end_Time,  
       sgr.notes  
FROM surgeryrecord sgr  
JOIN patients p ON p.patient_Id = sgr.patient_Id  
JOIN doctor d ON d.doct_Id = sgr.surgeon_Id  
JOIN nurse n ON n.nurse_Id = sgr.nurse_Id  
JOIN helpers h ON h.helper_Id = sgr.helper_Id  
WHERE  
       sgr.surgery_Date = '2024-05-16'  
       AND  
       sgr.patient_Id = 967  
       AND  
       sgr.start_Time = '23:15:52'  
       AND  
       sgr.end_Time = '23:45:52';
```

ID	Patient	Doctor	Nurse	Helper	Surgery Type	Surgery Date	Start_T	End_T	notes
967	Ammar	Mehboob	Wardah	Kishwar	Bariatric	2024-05-16	23:15:52	23:45:52	Transfer to icu

Question 12

The management wants to pay salaries for this month and wants a record of working hours. Each staff member should have 200 hours this month. For the month of May 2025, calculate the total working hours of each staff member (doctors, nurses, helpers) to check total hours according to the 200 hours baseline.

```
SELECT d.doct_id AS Staff_Id,
       d.fname AS Staff_Name,
       'Doctor' AS Staff_Type,

SUM(CASE
    WHEN sft.shift_End < sft.shift_Start THEN TIMESTAMPDIFF(HOUR, sft.shift_Start,
sft.shift_End) + 24
    ELSE TIMESTAMPDIFF(HOUR, sft.shift_Start, sft.shift_End) END) AS Total_Hours

FROM staffshift sft
JOIN doctor d ON d.doct_Id = sft.doct_Id
WHERE sft.shift_Date BETWEEN '2025-05-01' AND '2025-05-31'
GROUP BY
    d.doct_id,
    d.fname
```

UNION ALL

```
SELECT n.nurse_Id AS Staff_Id,
       n.fname AS Staff_Name,
       'Nurse' AS Staff_Type,

SUM(CASE
    WHEN sft.shift_End < sft.shift_Start THEN TIMESTAMPDIFF(HOUR, sft.shift_Start,
sft.shift_End) + 24
    ELSE TIMESTAMPDIFF(HOUR, sft.shift_Start, sft.shift_End) END) AS Total_Hours

FROM staffshift sft
JOIN nurse n ON n.nurse_Id = sft.nurse_Id
WHERE sft.shift_Date BETWEEN '2025-05-01' AND '2025-05-31'
```

GROUP BY

n.nurse_Id,
n.fname

UNION ALL

SELECT h.helper_Id AS Staff_Id,
h.fname AS Staff_Name,
'Helper' AS Staff_Type,

SUM(CASE
WHEN sft.shift_End < sft.shift_Start THEN TIMESTAMPDIFF(HOUR, sft.shift_Start,
sft.shift_End) + 24
ELSE TIMESTAMPDIFF(HOUR, sft.shift_Start, sft.shift_End) END) AS Total_Hours

FROM staffshift sft
JOIN helpers h ON h.helper_Id = sft.helper_Id
WHERE sft.shift_Date BETWEEN '2025-05-01' AND '2025-05-31'
GROUP BY
h.helper_Id,
h.fname;

Staff Id	Staff Name	Staff Type	Total Hours
1147	Dr. Nabeel	Doctor	28
1149	Dr. Imran	Doctor	2
1148	Dr. Rameez	Doctor	16
1150	Dr. Abu	Doctor	8
1156	Dr. Muhammad	Doctor	16

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Question 13

List all patients who have a follow-up appointment due this week, based on their last next_Visit from MedicalRecord.

```
SELECT p.patient_id AS Patient_Id,  
       p.fname AS Patient,  
       mdr.visit_Date AS Visit_Date  
  
FROM medicalrecord mdr  
JOIN patients p ON p.patient_Id = mdr.patient_Id  
WHERE  
       mdr.next_Visit IS NOT NULL  
       AND  
       mdr.next_Visit BETWEEN '2025-05-03' AND '2025-05-28';
```

Patient_Id	Patient	Visit Date
1037	Bushra	2025-02-06
570	Kabir	2025-02-06
1309	Rayyan	2025-02-06
1332	Kamil	2025-02-06
1266	Marwa	2024-07-10

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Question 14

Find the most preferred payment method chosen by upper-class people (defined by users of super deluxe room types).

```
SELECT rom.mode_of_payment AS Mode_Of_Payment,  
       COUNT(*) AS Usage_Count  
FROM roomrecords rom  
JOIN room r ON r.room_No = rom.room_no
```

```

JOIN patients p ON p.patient_Id = rom.patient_Id
JOIN appointment a ON a.patient_Id = p.patient_Id
JOIN doctor d ON d.doct_Id = a.doct_Id
WHERE
    r.room_Type = 'Super Deluxe Room'
GROUP BY
    rom.mode_of_payment;

```

Mode Of Payment	Usage Count
Card	61
Cash	46
Insurance	99

Question 15

The hospital wants to analyze the performance of its surgeons' surgeries. Give the percentage of stable patients as per declared in the notes after surgery.

```

SELECT d.doct_Id AS Surgeon_Id,
       d.fname AS Surgeons,
       COUNT(*) AS Total_Surgries,

SUM(CASE
    WHEN LOWER(sgr.notes) LIKE '%Stable%' THEN 1 ELSE 0 END) AS Stable_Patients,

ROUND(CAST(SUM(CASE
    WHEN LOWER(sgr.notes) LIKE '%Stable%' THEN 1 ELSE 0 END) AS FLOAT)
/
COUNT(sgr.surgery_Id) * 100, 2) AS Performance

FROM surgeryrecord sgr
JOIN doctor d ON d.doct_Id = sgr.surgeon_Id
GROUP BY
    d.doct_Id,
    d.fname;

```

Surgeon Id	Surgeons	Total Surgries	Stable Patients	Performance
1164	Dr. Akbar	30	7	23.33
1166	Dr. Bilal	33	5	15.15
1160	Dr. Rab	32	8	25
1171	Dr. Faisal	32	6	18.75
1157	Dr. Muhammad	22	4	18.18

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Thank You !!